



Medical RAG Assistant

An advanced Retrieval-Augmented Generation system for medical queries.

Project Overview

The Medical RAG Assistant answers medical queries on drug-drug interactions, food interactions, and general drug information.



Drug Interactions

Analyzes how different medications interact.



Food Interactions

Identifies potential interactions with food.

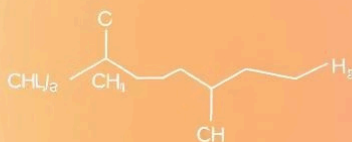
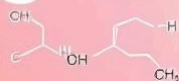


General Drug Info

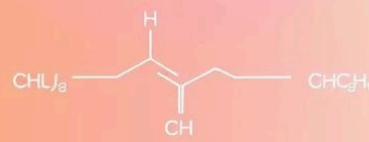
Provides comprehensive drug-related knowledge.

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Drug - Drug Interaction



DRUG			DRGK	
10cm	10em	Call	10cm	10cm
10.70	195%	451	10.30	159%
10.00	109%		10.9%	100%
285%	127%	522	10.9%	220%



ATT			DATA	
10cm	10cm	Call	10cm	10cm
50.1	10.997	59%	12600	220%
03%	10.9%	39.1	109%	138%
			109%	227%

key components:Dataset DrugBank Drug-Drug Interaction Dataset

- Description:** This dataset is manually sourced from FDA/Health Canada drug labels and primary literature, containing 191,808 DDI pairs involving 1,706 drugs. It includes 86 interaction types and is used for multi-class classification tasks to predict interactions based on the SMILES strings of drugs.

Key Components: RAG Pipeline

The RAG Pipeline retrieves context and supports multi-turn conversations.

Context Retrieval

Retrieves context from FAISS indices (drug, food, DDI datasets).

Query Expansion

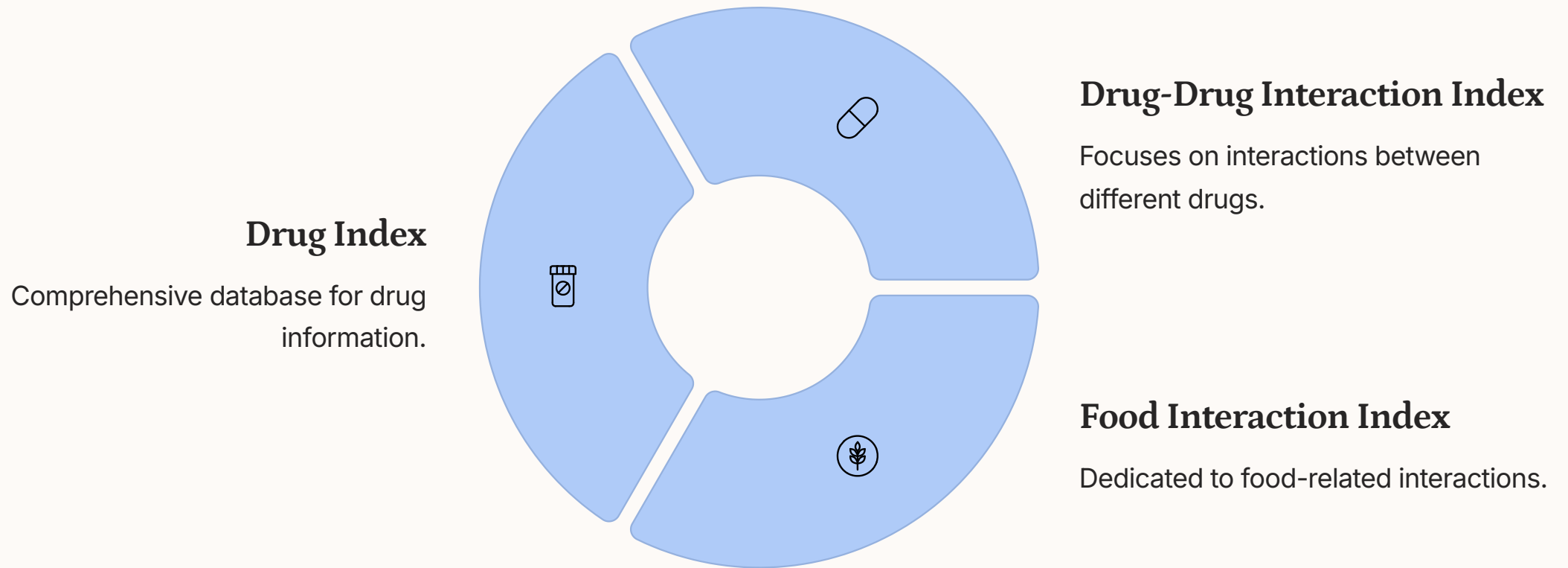
Expands queries using a synonym map for better results.

Multi-Turn Conversation

Supports ongoing dialogue through ChatManager.

Key Components: FAISS Index Architecture

Specialized indices ensure efficient and accurate information retrieval.



Key Components: Web Search AI Agent

Overview



The Web Search AI Agent significantly enhances the Medical RAG Assistant by integrating real-time, external web information. This is crucial when the primary FAISS database does not contain sufficient or updated medical evidence to answer complex queries.

How it Works: 7-Step Process



1. **User Query:** Patient or clinician submits a medical question.
2. **FAISS Check:** Initial retrieval from the internal FAISS knowledge base.
3. **Evidence Gap Detection:** Agent identifies if FAISS data is insufficient.
4. **Web Search Activation:** Initiates targeted web queries using specialized search algorithms.
5. **Intelligent Filtering:** Employs cosine similarity and medical relevance checks to filter results.
6. **Data Integration:** Combines validated web information with FAISS insights.
7. **Grounded Response:** Generates a comprehensive, medically accurate, and evidence-backed answer.

Key Features



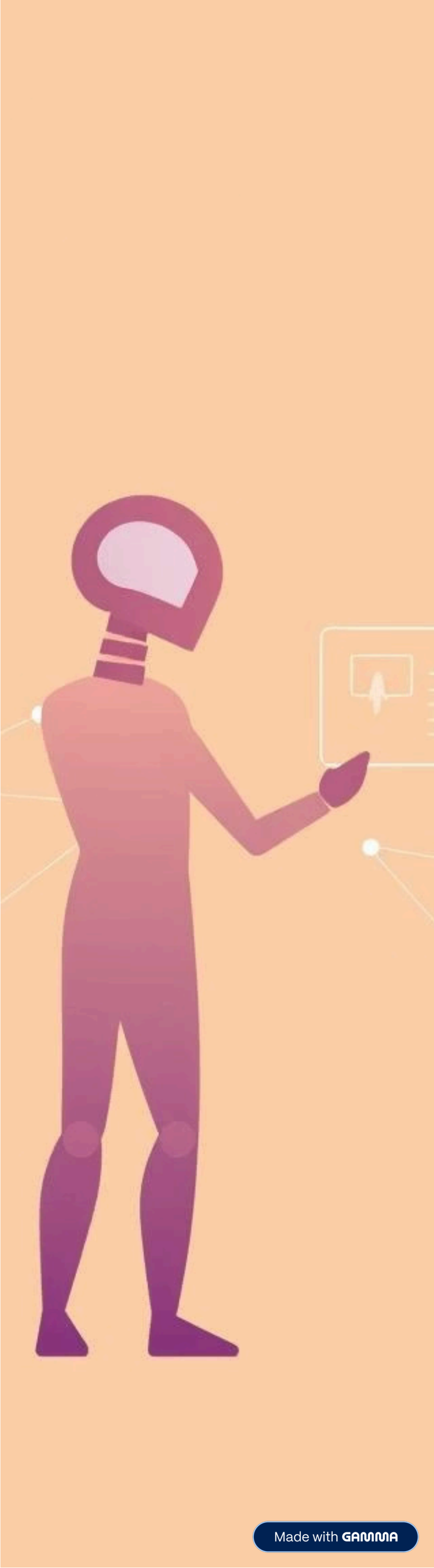
- **Cosine Similarity Filtering:** Ensures retrieved web content is highly relevant.
- **Seamless RAG Integration:** Smoothly merges real-time data with existing knowledge.
- **Safety Compliance:** Adheres to medical information retrieval guidelines, prioritizing patient safety.

Importance



This agent is vital for:

- **Preventing Hallucination:** Reduces the risk of AI generating incorrect or baseless medical information.
- **Adding Validated Information:** Incorporates external, up-to-date, and verified medical sources.
- **Enhancing Accuracy:** Improves the overall precision and reliability of the medical RAG assistant's responses.



Key Components: LLM Integration & Chat Memory

LLM Integration

- Uses Qwen or compatible causal LLM.
- Implements strict medical safety rules.
- Filters web search results via semantic similarity.

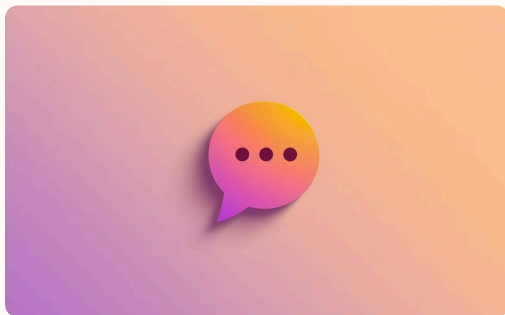
Chat Memory

- Stores multi-session histories.
- Summarizes and formats data for LLM context.



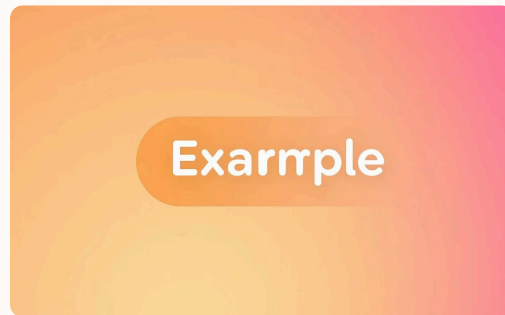
Key Components: UI Layer

A user-friendly Gradio interface for seamless interaction.



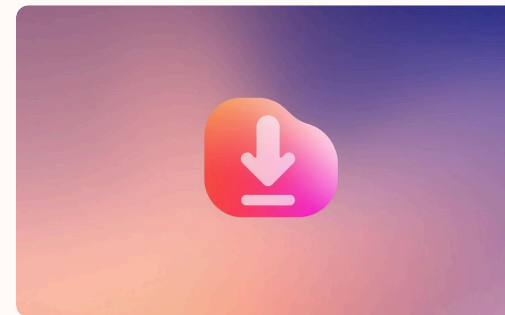
Chat Functionality

Intuitive chat for medical queries.



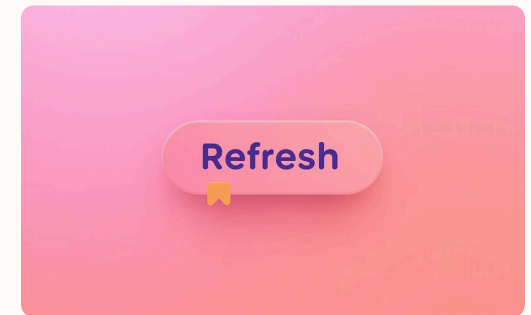
Examples

Pre-defined examples for guidance.



Export Options

Ability to export conversation history.



Reset Feature

Clear chat and start fresh.

Execution Requirements

The project relies on a robust set of modern AI and machine learning libraries.



PyTorch



Transformers



LangChain



FAISS CPU



Sentence Transformers

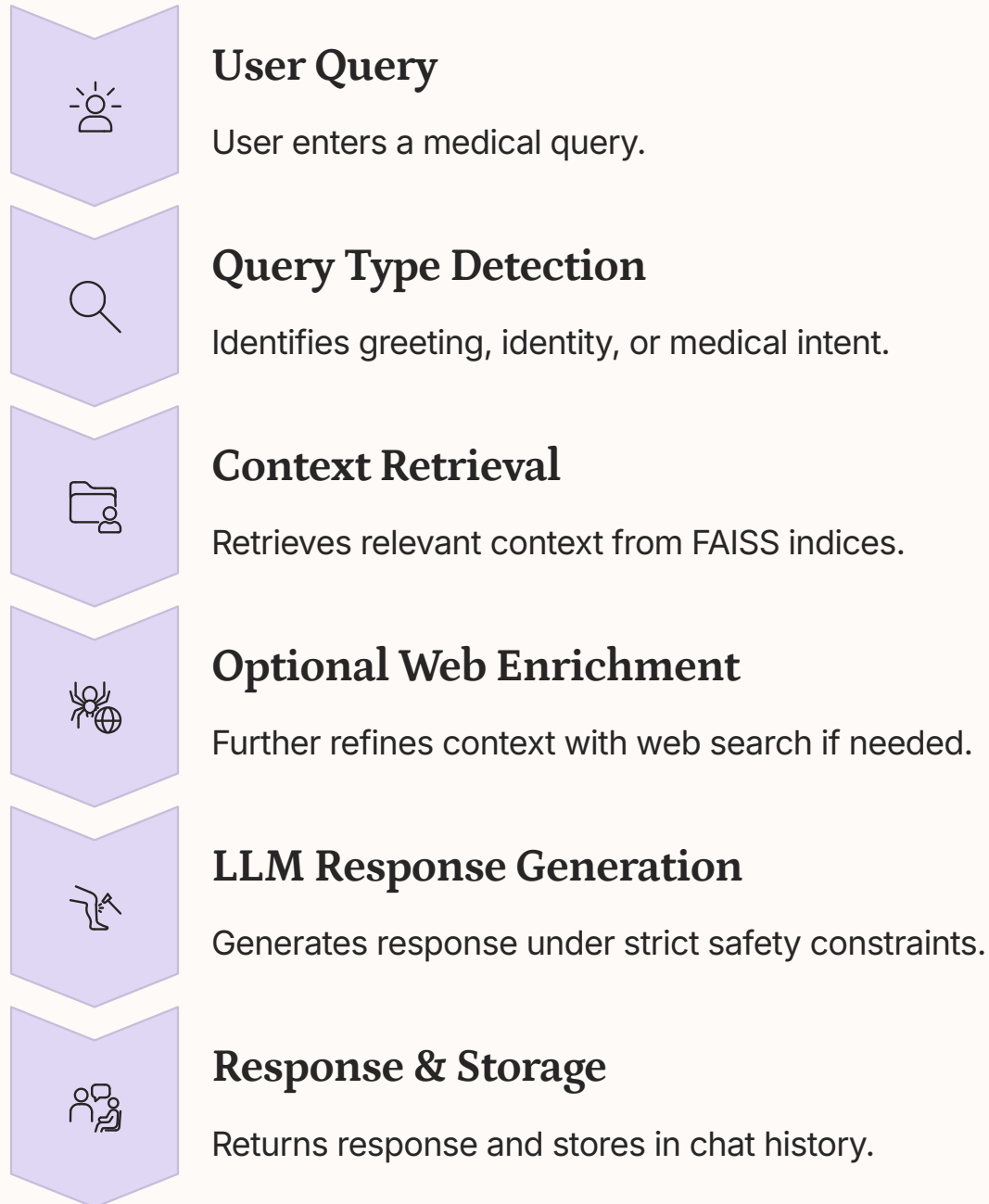


Gradio UI

Full requirements are detailed in the requirements.txt file.

System Flow: Query to Response

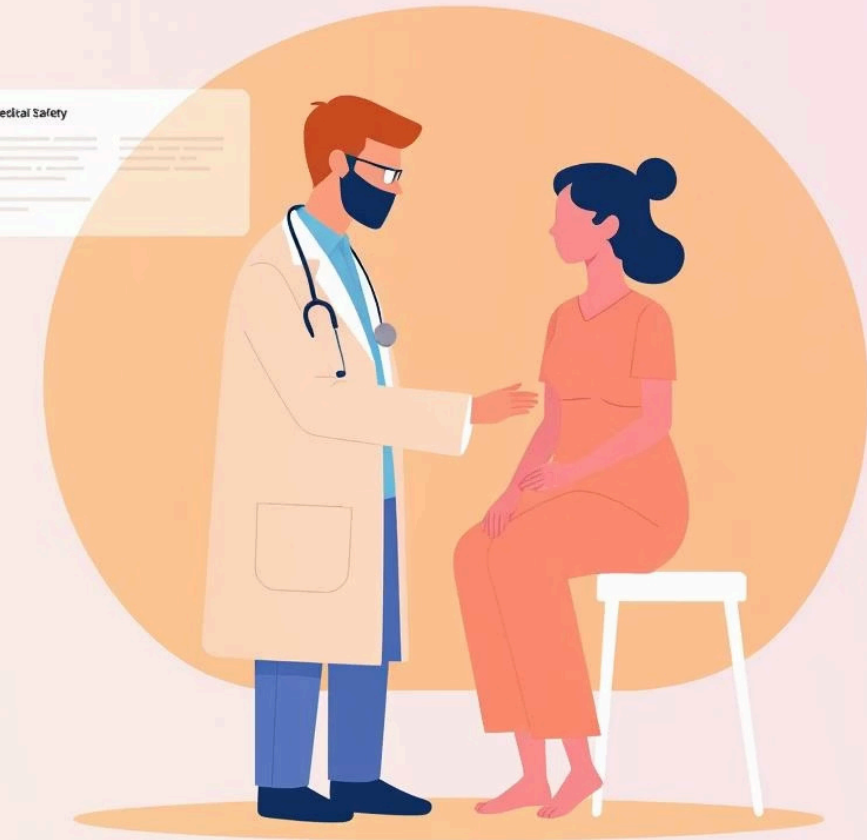
A step-by-step process from user query to a context-grounded medical response.



Safety Rules: Ensuring Responsible AI

Strict guidelines to prevent misinformation and promote responsible medical assistance.

- **No Hallucinations:** Avoid unsupported interactions or fabricated information.
- **No Treatment Recommendations:** Do not suggest dosages or treatment plans.
- **Context-Only Responses:** Use only provided context for answers.
- **Consult Professionals:** Always encourage users to consult healthcare providers.





Medical Assistant: Chat-Wise Intelligence

Our Medical Assistant leverages advanced AI to provide a highly intelligent and trustworthy conversational experience.

- **Intelligent Conversations:** Engages in natural language dialogues for a wide range of medical queries.
- **Context-Aware Responses:** Delivers relevant information by building upon previous interactions and understanding conversation flow.
- **Wise Decision-Making:** Employs sophisticated algorithms for accurate and thoughtful medical information retrieval.
- **Safe & Reliable Guidance:** Prioritizes accuracy and safety, offering trustworthy responses to healthcare-related questions.
- **Multi-Turn Memory:** Supports seamless, extended conversations with robust memory retention for continuity.

Thank You

Team NHA-290

Lujan Hussam

Muhammed Mostafa

Omnia Mahmoud

Zyad Yasser

Noureen Muhammed